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Module 4: Software Design
Session 3: Use Cases, User Stories and Test Scenarios

Software Developer Level 4

What you will be able to do

By the end of this session you will be able to:

1. Construct a use case with a main flow and alternative flows (extensions) from a business scenario.
2. Write user stories that meet INVEST, with Given/When/Then acceptance criteria that cover boundaries and safety rules.
3. Derive test scenarios from acceptance criteria and trace each one to its requirement.

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From your preparation

WARM-UP

Which rule in the Repair Priority Policy would be easiest to get wrong in code?

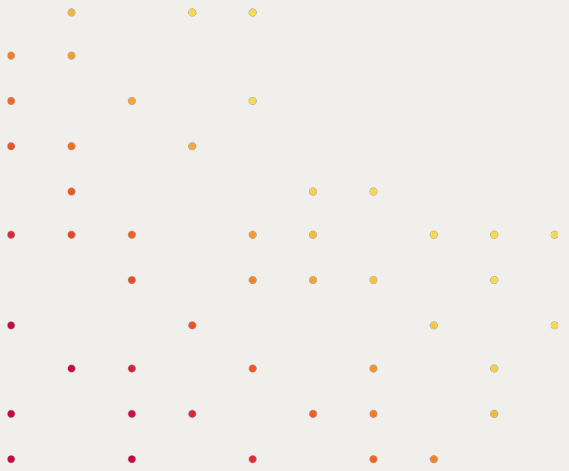
Type the rule and your reason in the chat.

Then, from memory: the three parts of a user story, and of a Given/When/Then criterion.

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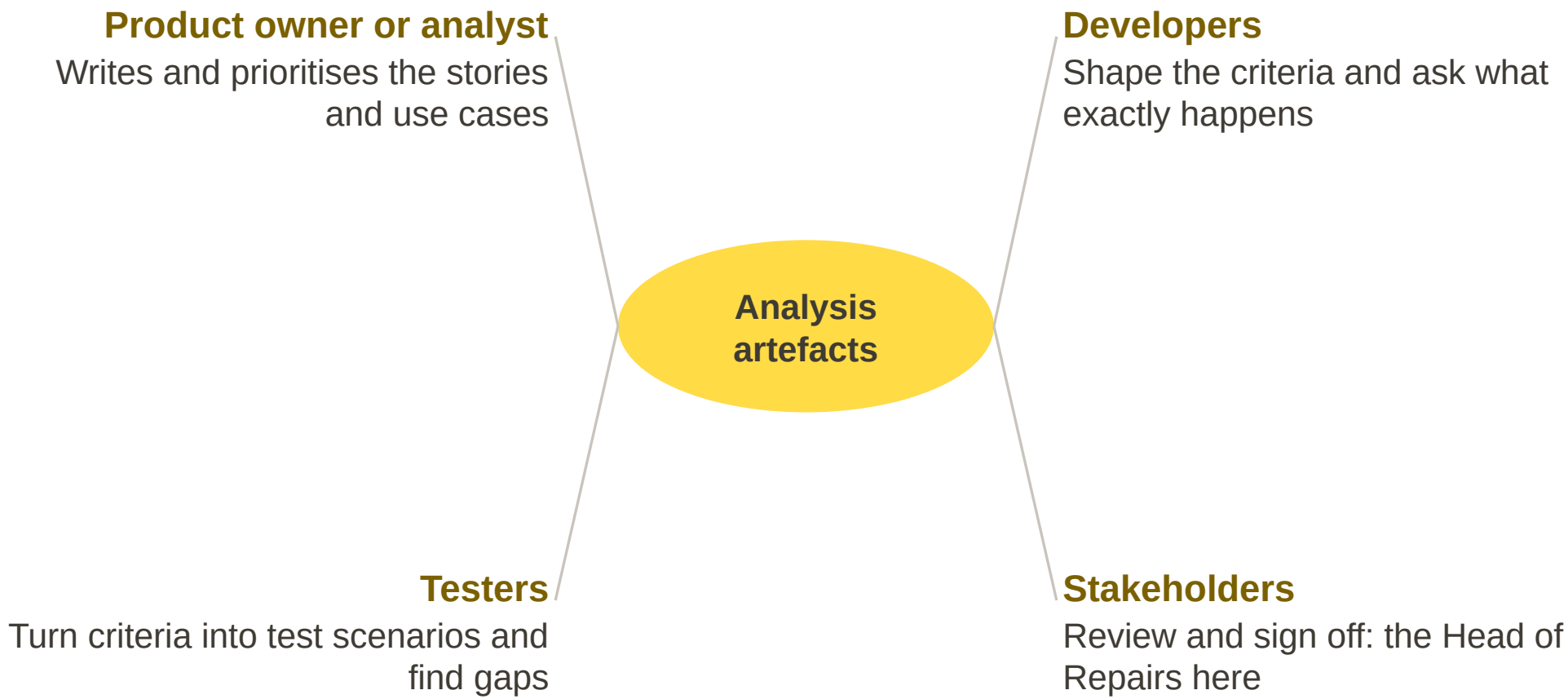
Who writes what

Who owns each analysis artefact, and why a use case counts as a specification.



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Who writes, shapes, tests and signs off



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The conversation before the build

Before a story enters a sprint, three views read it together.

Product owner: is this what the business needs?

Developer: what exactly happens, with what data?

Tester: how will we know it works, and what could go wrong?

Kestrel Moor's four developers work with an agency, so the answers must be written down.



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A use case is a specification

FUNCTIONAL SPECIFICATION

> **A use case specifies what the system must do so that one actor reaches one goal.**

It names the actor, the goal, the normal path and every way it can go wrong.

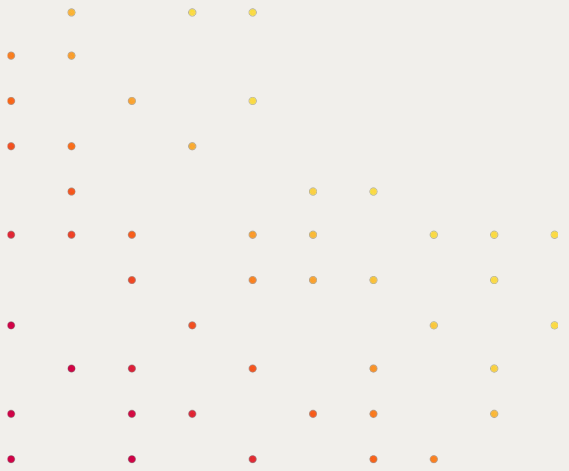
It leaves out screens, buttons and code, so it stays true while the design changes.

Developers build to it, testers derive tests from it, stakeholders can read it.

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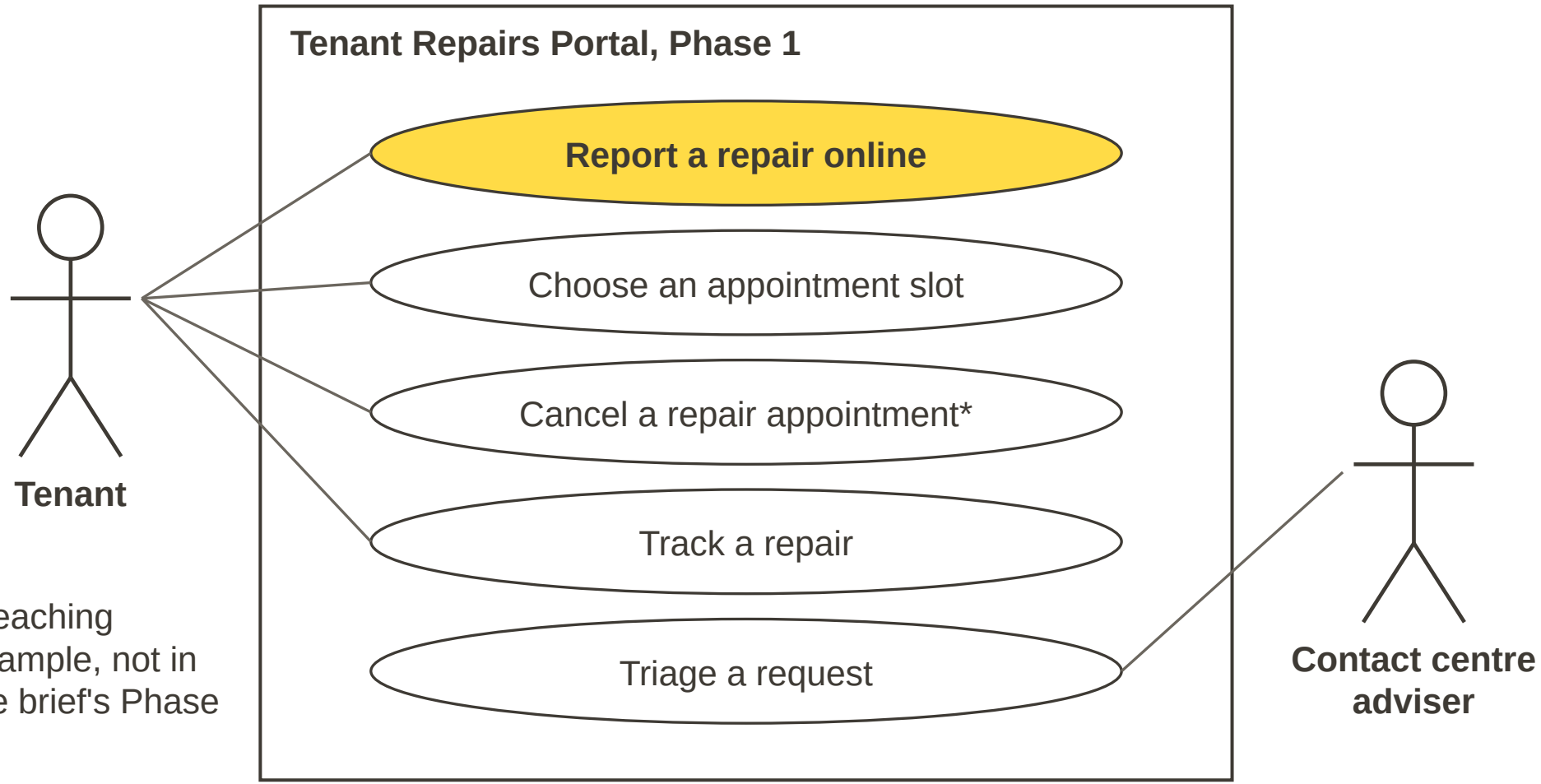
Use cases

Actor, goal, trigger, the main flow, alternative flows (extensions) and conditions.



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The portal's use cases on one page



*Teaching example, not in the brief's Phase 1

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Worked example: Cancel a repair appointment

USE CASE

(1) Actor: Tenant

Goal: cancel a booked visit and free the slot

(2) Preconditions: signed in; a visit is booked

Trigger: tenant chooses Cancel on the visit

(3) 1. Tenant opens the repair and chooses Cancel.

2. System shows the date, time and trade.

3. Tenant gives a reason and confirms.

4. System cancels the slot with the scheduler.

5. System confirms by the tenant's chosen channel.

(4) Postconditions: slot freed; the operative's job list updated

1. One primary actor and a goal in the actor's words.

2. Already true at the start, so the steps do not check them.

3. Steps alternate actor and system and move towards the goal.

4. What is true when the goal succeeds.

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Worked example: alternative flows (extensions)

Branches from	Condition the system detects	What the system does
Step 2 (2a)	The operative is already on the way	Says it is too late to cancel online and shows the contact centre number; use case ends
Step 3 (3a)	Tenant wants a new date instead	Offers 'Choose an appointment slot', a separate use case
Step 4 (4a)	The scheduler does not confirm	Keeps the cancellation as pending, tells the tenant and confirms by text when done

> **Same goal, different path: an extension. A different goal: a separate use case.**

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Write steps the actor would recognise

ONE RULE FOR STEPS

> **Each step is an actor passing information, the system checking something, or the system recording something.**

Weak: 'System calls DELETE /slots/{id}.' Strong: 'System cancels the slot with the scheduler.'

Weak: 'Tenant fills in the form.' It hides every decision the tenant makes.

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Activity 1: Write the use case

Pairs (breakout rooms)

Specify the online repair report as a use case.

1. Alone (2 min): read KMR-26-0019, 0039 and 0115 (Supporting Material s2); tick any needing a branch.
2. In workbook Part 1: write actor, goal and trigger.
3. Write the main flow: at least six steps, each one tenant or system action.
4. Write the emergency extension: branching step, condition and what the system does.
5. Request check: the step or extension for KMR-26-0019, KMR-26-0039 and KMR-26-0115.

Extension: see the Activities document

TIME
24 minutes working time

GROUPING
Pairs (breakout rooms)

YOU NEED
Analysis workbook, Part 1
Supporting Material, section 2
Worked example: section 1

OUTPUT AND SUCCESS
Core: actor, goal, trigger, six or more main steps, the emergency extension and three request checks.

Debrief: Which request did your first main flow miss, and what did you add?

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Break

20 minutes

Check your pair's workbook is saved in your own OneDrive. After the break you work on your own from it.

User stories and acceptance criteria

INVEST, Given/When/Then, boundaries and safety rules as criteria.

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INVEST: a checklist for a good story

Letter	Quality	Ask
I	Independent	Can it be built and released without waiting for another story?
N	Negotiable	Does it leave the how open for the team to discuss?
V	Valuable	Would a tenant or Kestrel Moor notice the benefit?
E	Estimable	Could the team give it a rough size today?
S	Small	Can it be finished inside one sprint?
T	Testable	Could a tester write a test from it without asking you?

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Worked example: a story and its criteria

RENT SYSTEM (NOT THE PORTAL)

(1) As a tenant who pays rent by card,
I want a reminder when rent is due in 3 days or fewer,
so that the money is ready on the day.
(2) AC1 Given rent due Mon 9 Nov is unpaid and no reminder has gone, When the daily job runs on Fri 6 Nov, Then one reminder goes by the tenant's chosen channel.
(3) AC2 Given the same, When it runs on Thu 5 Nov, Then no reminder is sent.
(4) AC3 Given rent due Mon 9 Nov is already paid, When the job runs on Fri 6 Nov, Then no reminder is sent.

1. Role, need and benefit. No channel or screen named.
2. On the boundary: 3 days before. Given holds the data.
3. Just outside: 4 days before. Both sides matter.
4. Negative: the condition is false, so nothing happens.

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Boundaries: test on the line and either side of it

Rule: an unpaid tenant gets one reminder when rent is due in 3 days or fewer.



Just outside
4 days before: no reminder

On the boundary
3 days before: reminder sent

Just inside (optional)
2 days before: sent if none has gone

Due date
Rent is due

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Safety rules belong in the criteria

4 of 4

written reports of gas or burning electrics were triaged Routine

2 gas smells and 2 sparking or burning electrics came by web form or email. All were given Routine.

All 5 phone reports of the same kinds were given Emergency.

An adviser applies the policy on the call. A form only does what its criteria say.

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Activity 2: Stories and acceptance criteria

Individual, in breakout rooms

Turn the use case and policy into testable stories.

1. In workbook Part 2, write two user stories from your pair's use case.
2. Write a boundary pair from the policy: one criterion on the limit, one just outside it.
3. Write a safety criterion for one emergency, and a repeat damp criterion.
4. Write each as Given/When/Then with an ID (AC1 to AC4); list each story's IDs.
5. INVEST check: for each story, name its weakest letter and write your fix.

Extension: see the Activities document

TIME
25 minutes working time

GROUPING
Individual, in breakout rooms

YOU NEED
Analysis workbook, Part 2
Supporting Material, section 3
Policy extract: section 2

OUTPUT AND SUCCESS
Core: two stories, AC1 to AC4 (boundary pair, safety, repeat damp) and an INVEST fix for each story.

Debrief: Read out your boundary pair: which date or hour sits on each side?

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From criteria to test scenarios

Positive, negative and boundary tests, each traced to the criterion it checks.

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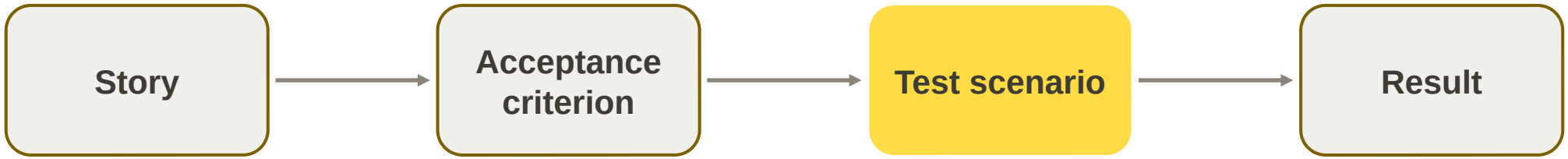
Worked example: tests from the rent criteria

Test ID	Criterion it checks	Input or data	Expected result
T1	AC1 (on the boundary)	Unpaid; due Mon 9 Nov; job runs Fri 6 Nov	One reminder by the chosen channel
T2	AC2 (just outside)	Unpaid; due Mon 9 Nov; job runs Thu 5 Nov	No reminder
T3	AC3 (negative)	Paid; due Mon 9 Nov; job runs Fri 6 Nov	No reminder
T4	None covers it	Unpaid, no reminder yet; job runs Sat 7 Nov	Unknown: ask the product owner

Positive, negative and both sides of each boundary; one criterion per test.

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Traceability: from story to test result



Each story lists its criteria

Each criterion has at least one test

A failed test names its criterion

Follow the IDs back from a failed test to the story to reopen and the person to ask.

Before you test

Could a tester write a test from this criterion without asking you?

If not, the criterion is not finished.

Next, you apply this question to your partner's criteria.

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Activity 3: Swap and test

Pairs (breakout rooms)

Prove your partner's criteria can be tested.

1. Swap: post your workbook in the breakout chat and save your partner's copy.
2. In your own workbook Part 3, write four tests for your partner's criteria: a positive test on their boundary, one just outside it, one for their safety rule and one negative.
3. Give each test the criterion ID it checks, the exact input or data, and the expected result.

Extension: see the Activities document

TIME
18 minutes working time

GROUPING
Pairs (breakout rooms)

YOU NEED
Your partner's Part 2
Analysis workbook, Part 3
Supporting Material, section 4

OUTPUT AND SUCCESS
Core: four traced tests: on the boundary, just outside it, the safety rule and one negative.

Debrief: Which criterion was hardest to test, and what made it hard?

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Check your understanding

CHECK

1. In refinement, a criterion says 'Then the tenant is told promptly'. What should happen?

A The team asks the product owner to agree an observable Then
B The developer picks a time limit while coding the story
C The tester decides what promptly means when testing it
D The Head of Repairs rewrites the story at design sign-off

2. No recorded vulnerability: which test pair checks the 1 October boundary?

A Reports on 1 October and again on 15 October
B Reports on 1 October and on 30 April
C Two reports on 1 October, one by phone
D Reports on 30 September and on 1 October

3. In 'Report a repair online', which is an extension, not a separate use case?

A Tenant cancels an appointment that was booked last week
B Adviser triages the web queue at 09:00
C Photo upload fails; tenant carries on without photos
D Tenant checks the status of an earlier repair

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Skills covered today

KSB and official wording	In plain English	Where we covered it today
Part of K2: roles and responsibilities within the software development lifecycle (who is responsible for what)	Who writes, uses and signs off each analysis artefact	Who writes what; Activity 3; live check
Part of K11: functional or technical specifications	Using a use case as a specification of what the system must do	Activity 1; live check
S6: identify and create test scenarios	Turning acceptance criteria into tests, including boundaries	Activity 3; live check
S9: create analysis artefacts, such as use cases and/or user stories	Writing a use case, user stories and acceptance criteria	Activities 1 and 2
Part of B5: safety	Making sure the portal stops an emergency report and sends every tenant, including those who cannot phone, to the right number	Activities 1 to 3

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What to take away

1. A use case specifies one goal: a main flow and alternative flows (extensions).
2. A story should pass INVEST; every Then must be something a tester can see.
3. Every boundary needs a criterion and a test on each side of it.
4. An emergency cannot be submitted online: the portal shows the numbers to call, including the text relay route, and no Submit.
5. Each test names its criterion, so a failure points to a requirement.

Before Session 4

Do the Session 3 Consolidate Quiz (about 10 minutes, answers on the last page).

Complete the Prepare document for Session 4: Architecture, UML and Workflow Modelling.

Keep your workbook: Session 4 starts from your use case.

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Reflective Log: your Session 3 entry

Complete the Session 3 entry in Reflective Log.docx now (about 5 minutes).

Prompt: one boundary acceptance criterion and the test scenario that checks it.

Then: how this could help in your own work.

Copy the key points into your OneFile timesheet reflection.

Keep workplace examples general: no personal or customer data, credentials or confidential material, and no employer data in unapproved AI tools.

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